SUBJECT	Lower Secondary Science Year 8
TOPIC	The Periodic Table & Separation Techniques
ASSIGNMENT	Two
STUDENT NAME	

FEEDBACK AND REFLECTION:

- I have read the feedback on my last assignment carefully.
- I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: ____

____

____

____

ASSIGNMENT GUIDANCE:

- You should complete this assignment by carrying out Practical 1: Solubility on this sheet and then handwriting in black ink both the Practical 1: Solubility questions and the assignment questions that follow on, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 1 and 2 hours for the practical, and between 40 and 60 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment Two

PRACTICAL 1: SOLUBILITY

How does the temperature of water affect how much sugar can dissolve in it?

Apparatus

- Sugar
- Kettle
- Insulated mug – a thermos or travelling mug would be good, if you can see into it and stir it. If you do not have one of these, a polystyrene cup, or a mug wrapped in cloth would also work.
- Measuring jug that can hold hot water
- Kitchen scales or electric balance
- 2 spoons – one for stirring, one for adding sugar. If you use the same one your sugar will get wet, and it could affect the results … but it will also probably irritate the person who wants to use the sugar next!
- Thermometer
- A responsible person to supervise the use of boiling water!

Method

Read the whole thing through before you start.

First of all, keep in mind that you probably will not be able to get the temperatures exactly right, so rather than choosing particular temperatures to put into your table, you will just have to use different proportions of hot and cold water, and measure what temperature you get.

- Use your measuring jug to add 100ml of tap water into your cup. Because of the density of water, this will have a mass of 100g. Record the temperature of the water in your table. Make sure you give the thermometer time to settle at the correct temperature, do not try and read it while the temperature is changing rapidly!

- Put your stirring spoon in the cup, place your cup on the scales, and set them to zero. This means only the mass of the sugar you add will be counted.

- Add sugar by the spoonful, stirring all the time, until you start to notice that there are crystals that will not dissolve. Record the mass in your table.

- Repeat the experiment with boiling water. (**Please be careful!** Also note that this will cool down quickly if your cup is not insulated. That will affect your results, so try and do it quickly, but always choose to be safe first.)

- Repeat the experiment with 50ml of cold water, and 50ml of hot water. Make sure you stir this well before taking the temperature readings.

- You could repeat it again with a different ratio of cold and hot water, which will give you a different temperature again.

1. This question covers the practical.

Variables

a. What is your independent variable in this experiment?

____

(1 mark)

b. What is your dependent variable in this experiment?

____

(1 mark)

c. Nothing should change in an experiment except for the independent and dependent variables. Everything that could possibly affect the dependent variable should be controlled (kept the same). What do you need to keep the same (control) in this experiment?

____

____

____

(2 marks)

d. What kind of thing might scientists in a high-tech lab be able to control or do differently that you cannot at home?

____

____

(1 mark)

Results

e. First complete the headings of your table, (**Hint** - look at the tables on page 113 if you are not sure what the headings should be) then fill in your results.

(2 marks)

f. Now draw a graph of your results using crosses and a line of best fit.

- Put the temperature on the horizontal axis and the solubility on the vertical axis.
- Do not forget to use a regular scale on both axes (go up in 1cm, or 2cm, or 5cm increments, do not just go up in random values). Make sure you use a scale that stretches out so you use as much of the graph space as you can.
- Plot your points carefully, and draw a line of best fit, do not 'join the dots'.

(4 marks)

(Total 11 marks)

2.

a. Draw a straight line between the **name** of the element and its **chemical symbol.**

name	chemical symbol
	Al
neon	Na
sodium	Ar
nitrogen	Ne
aluminium	N
	S

(4 marks)

b. Circle the **two** elements that are metals.

aluminium **neon** **nitrogen** **sodium**

(2 marks)

c. Give **three** properties of metals.

1. ______________________________

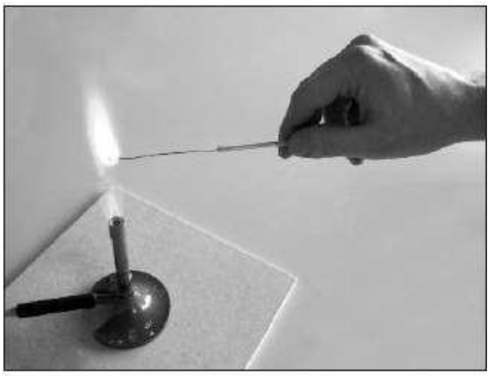
2. ______________________________

3. ______________________________

(3 marks)

(Total 9 marks)

3. When copper metal is heated it reacts with a gas in air.



a. What is the chemical name of the **product** formed when copper reacts with a gas in air?

____

(1 mark)

b. Which statement below describes what happens in a **chemical change** but **not** in a physical change? Tick the correct box.

The product is a solid.	☐
The change only happens at a high temperature.	☐
The atoms have combined in a different way to make a new substance.	☐
The types of atoms at the start are the same as in the end product.	☐

(1 mark)

(Total 2 marks)

4. Lithium is a group 1 metal. Some batteries contain lithium. Amy wants to find out what happens if air or water react with lithium.

   a. Amy cuts a fresh piece of lithium.

   The fresh surface of the lithium reacts with oxygen in the air.

   What does Amy see when the surface reacts with oxygen? Put a tick in the box next to the correct answer.

The surface bubbles and fizzes.	☐
A flame appears.	☐
The surface changes from shiny to dull.	☐
The piece of lithium gets smaller.	☐

   (1 mark)

   b. When lithium reacts with oxygen, **lithium oxide** is made. Write a word equation for this reaction.

   ____

   (2 marks)

   c. Amy drops a freshly cut piece of lithium into a beaker of water.

   What does Amy see when the lithium reacts with the water? Put ticks in the boxes next to the **two** correct answers.

The lithium moves around.	☐
The lithium sinks to the bottom of the water.	☐
The level of the water rises.	☐
The piece of lithium gets bigger.	☐
The lithium fizzes and bubbles form.	☐

   (2 marks)

d. A gas is made in the reaction. What is the name of the gas? Put a ring around the correct answer.

carbon dioxide **chlorine** **hydrogen** **oxygen** **nitrogen**

(1 mark)

The next two elements in group 1 are sodium and potassium.

Amy finds out some information about lithium, sodium and potassium.

element	**symbol**	**melting point in ℃**
lithium	Li	180
sodium	Na	
potassium	K	64

e. Which of the following is most likely to be the melting point for sodium? Put a ring around the correct answer.

40℃ **63℃** **97℃** **181℃** **200℃**

(1 mark)

f. Which of these elements reacts most **slowly** with water?

____

(1 mark)
(Total 8 marks)

5.

Chlorine, bromine and iodine are in Group 7 of the Periodic Table.

a. Draw three straight lines to connect each **element** to its correct **appearance** at room temperature and pressure.

element	appearance
	grey solid
	red-brown solid
chlorine	green gas
bromine	grey gas
iodine	red-brown liquid
	green liquid

(3 marks)

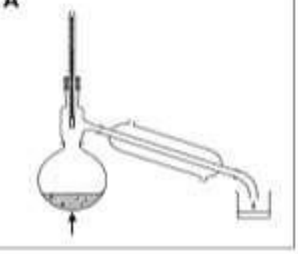
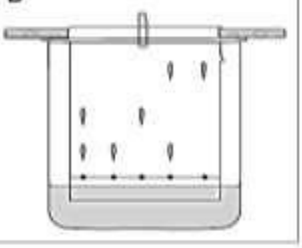
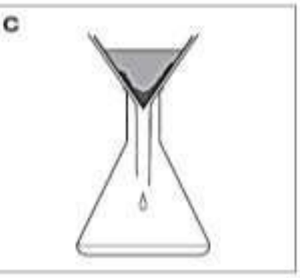
b. Joe does an experiment. He reacts sodium with chlorine to make sodium chloride. Write a **word equation** for this reaction.

____

(2 marks)
(Total 5 marks)

6. Diagrams A, B and C show three pieces of apparatus for separating substances.

a. Draw a line from each apparatus to the name of the method of separation.
Draw only **three** lines.

diagram of apparatus	method of separation
	chromatography
A 	
	distillation
B 	
	filtration
C 	
	crystallisation

(3 marks)

Debbie has a mixture of sand and salt water. Look at the diagrams above.

b. Which apparatus would Debbie use to separate the sand from the salt water? Give the correct letter.

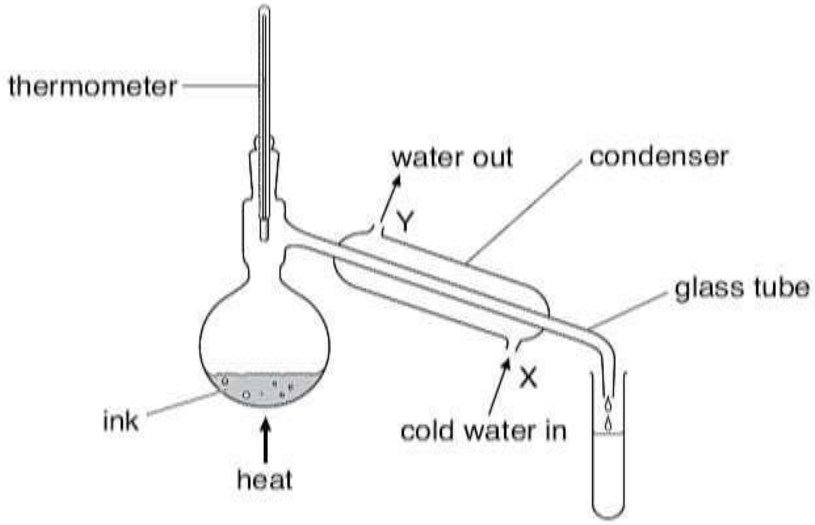
_________ (1 mark)

c. Which apparatus would she use to separate pure water from the salt water? Give the correct letter.

_________ (1 mark)

(Total 5 marks)

7. Rema used the apparatus below to distil 100cm³ of water-soluble ink.



apparatus A

a. Which processes occur during distillation? Tick the correct box.

condensation then evaporation	☐
evaporation then condensation	☐
melting then boiling	☐
melting then evaporation	☐

(1 mark)

b. Give the name of the colourless liquid that collects in the test tube.

____

(1 mark)

c. What would the temperature reading be on the thermometer when the ink has been boiling for two minutes?

_________°C

(1 mark)

d. Water at 15°C enters the condenser at X. Predict the temperature of the water when it leaves the condenser at Y.

________°C

Explain this change of temperature.

____

____

(1 mark)

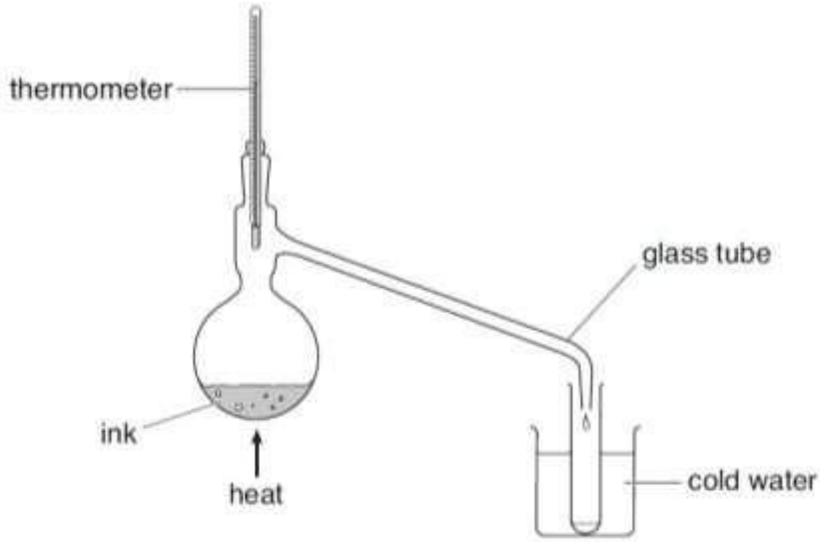
e. Give **two** ways in which the water vapour changes as it passes down the glass tube inside the condenser.

1. ____

2. ____

(2 marks)

f. Peter used the apparatus below to distil 100cm³ of water-soluble ink.



apparatus B

Why is the condenser in **apparatus A** better than the glass tube and beaker of water in **apparatus B**?

____

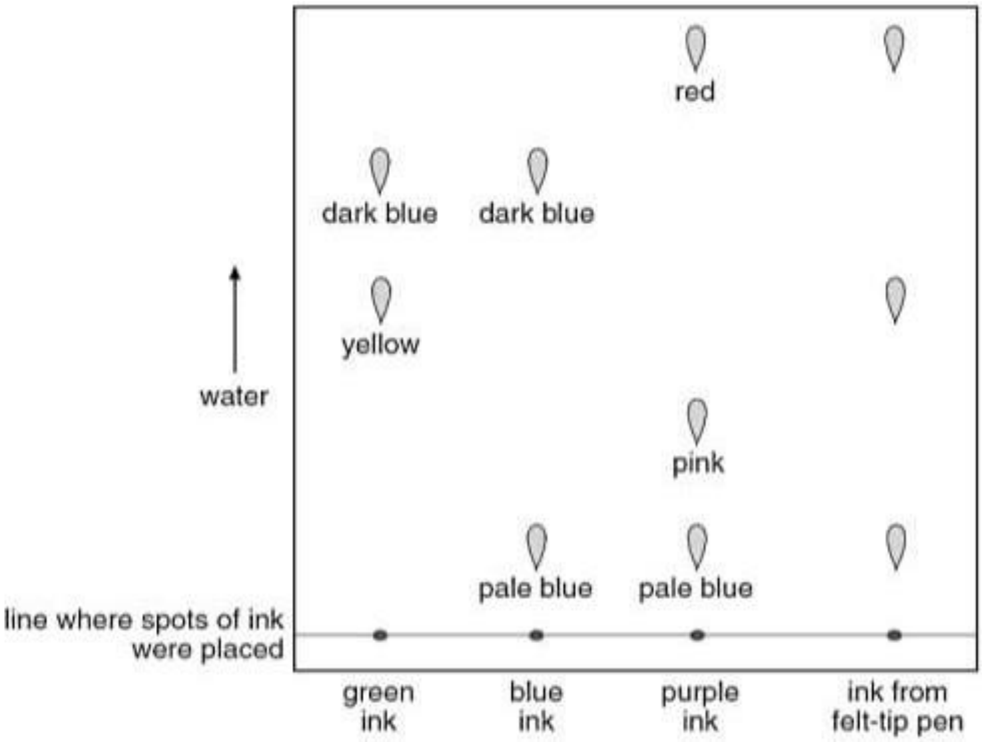
(1 mark)
(Total 7 marks)

8. Susie used chromatography to identify the coloured substances in the ink from a felt-tip pen.

She used:

- green ink
- blue ink
- purple ink
- ink from her felt-tip pen

She used water as the solvent.



Look at the diagram above.

a. Which colours were present in the ink from the felt-tip pen?

____

(1 mark)

b. How many coloured substances were there in green ink?

____

How can you tell?

____

____

(1 mark)

c. Susie placed the spots of ink on a line on the chromatography paper as shown in the diagram. To draw the line, Susie had to choose a felt tip or a pencil. Which **one** should she use?

____

Give the reason for your answer.

____

____

(1 mark)

d. Susie used water as the solvent in this experiment. When she repeated the experiment with a different set of pens, it did **not** work.

She then used ethanol instead of water. Suggest why the experiment worked with ethanol but **not** with water.

____

____

(1 mark)
(Total 4 marks)

TOTAL FOR ASSIGNMENT 51 MARKS